SolarGuard Project Report: Automated Solar Panel Cooling and Cleaning System

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This report compiles everything produced for the SolarGuard project so far: the design, the full pin-to-pin wiring, the 3D printed parts, the simulations, the related-work evidence and the work that remains. It is written so each section can be lifted into your project report.

1. Executive summary

SolarGuard is an ESP32-controlled system that cools a solar panel with a water spray when it overheats and cleans it with a wiper carriage when dust cuts its output, while an identical untouched panel acts as the control. So far the project exists as a complete design, simulations and generated 3D part files; nothing has been built or measured yet.

The problem. Dust (soiling) and high cell temperature both lower a panel's output. The literature reviewed in sections 3 and 4 reports dust losses of roughly 7 to 40 percent depending on climate and exposure time, and a power loss of about 0.4 to 0.45 percent for every degree Celsius above 25 °C. Existing student and research prototypes usually solve one of the two problems, or do both without measuring whether the pump and motor cost more energy than they save.

The design in one paragraph. Two 12 V, 20 W panels sit side by side. Panel A is cleaned and cooled; Panel B is the control. An ESP32-S2 reads panel voltage and current (ACS712 sensors and resistor dividers), three DS18B20 temperature probes, a DHT22, a small reference solar cell for sunlight, a rain probe, a tank level sensor and two carriage end stops. A relay switches a 12 V pump that sprays the panel from a fixed spray bar. An L298N drives a gear motor that moves a squeegee carriage down and back on two 8 mm rods. A 3-cell Li-ion pack with a BMS and a solar charge controller powers the electronics. A second relay can switch Panel A onto a dummy load so both panels are compared at identical load.

What has been produced.

- Component review, corrected parts list and an Excel tracker (68 line items).
- Circuit diagrams, a pin-to-pin netlist and a pin allocation for the ESP32-S2.
- 12 printable 3D parts (STL and OpenSCAD) including a fit-test coupon.

- A browser digital twin with thermal, electrical, soiling, water and battery models.
- A Wokwi project that runs the control logic on a simulated ESP32-S2.
- This related-work analysis of the four papers you supplied plus further sources.

What is proven and what is not. The control logic was tested on a software mock (boot, cooling, cleaning, rain, low tank, jam and sensor faults). The 3D meshes were checked for watertightness and hole sizes. Nothing electrical or mechanical has been tested on real hardware, no prices were verified, and no measured data exists yet. Section 16 lists the data that will turn the design into a result.

What remains. Buy the missing parts, print the coupon and parts, build and calibrate the sensing, run the A-versus-B experiments, compute net energy gain, run a patent search, and write the paper. Section 21 gives the roadmap.

2. Background: why dust and heat matter

Both dust and heat reduce a panel's output, and the two losses compound. Dust blocks and scatters light, so less current is generated. Heat lowers the cell voltage, so each watt of sunlight yields less power. A system that treats only one of them leaves part of the loss on the table.

2.1 Dust (soiling)

The numbers below are what published studies report. They vary by climate, dust type, tilt and exposure time, so use them as evidence that the effect exists, not as predictions for your site.

Reported finding	Value	Where it comes from
Cairo, 20 W panel, two weeks without cleaning	dirty-panel power about half of clean; cleaner restored about 99.4%	Anber and Aboshosha, 2024
Riyadh, 8 months of exposure	about 32% loss in panel performance	Mani and Pillai and Salim et al., as cited in Anber and Aboshosha
Kuwait City, 6 days of sand	about 17% drop in electricity	Wakim, as cited in Anber and Aboshosha

Reported finding	Value	Where it comes from
Bangladesh, 1 month	efficiency reduced by up to 35%	Rahman et al., as cited in Anber and Aboshosha
Taxila, Pakistan, 3 months in winter	power down about 20% (mono) and 16% (poly)	Ali et al., as cited in Anber and Aboshosha
India prototype, 1 day / 1 week / 1 month of dust	efficiency loss 7.32% / 10.78% / 14.52%	Balamurugan and Nithya, 2025
Athens, urban air pollution	up to about 6.5% less energy	Kaldellis and Kokala, as cited in Anber and Aboshosha

Two practical points follow. First, dust loss accumulates over days, so a trigger based on measured loss (rather than a fixed calendar) avoids both under- and over-cleaning. Second, the Cairo study recommends cleaning roughly every two weeks to hold efficiency near 90 percent, which is a useful sanity check for your own schedule.

2.2 Temperature

Crystalline silicon modules lose power as they heat above the 25 °C test condition. Skoplaki and Palyvos review the standard correlations for this effect (Solar Energy 83(5), 2009); a temperature coefficient near 0.4 percent per degree is the figure commonly used. The University of Sharjah inventors quote 0.45 percent per degree and note that panels can reach 70 °C, costing up to about 20 percent of output (press release).

This project uses 0.4 percent per degree in the digital twin and for the expected-gain calculation:

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P = P_{STC} \cdot \frac{G}{1000} \cdot SI \cdot \left[1 - \gamma \,(T_{cell} - 25)\right], \quad \gamma = 0.004\ \text{K}^{-1}
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Here G is irradiance in W/m2, SI is the soiling index (1 for a clean panel) and T_cell is the panel temperature in °C. For a 20 W panel at 55 °C and SI = 1, the temperature term alone gives 20 x (1 - 0.004 x 30) = 17.6 W, a loss of 2.4 W that cooling can partly win back.

2.3 Why a combined, measured approach

Cleaning also cools a panel slightly and cooling with a front spray also washes it, so the two functions interact. That is why the experiment in section 16 measures cooling-only, cleaning-only and combined runs against a control, and why it counts the energy the

pump and motor consume.

3. Related work, part 1: the four sources you listed

I opened every link. Two papers opened in full, one only as an abstract, and one is a press release about a patent. The table says exactly what was readable, so you know how far each source can be quoted.

#	Source	What was readable	Closest to which part of SolarGuard	Use as
i	Anber and Aboshosha, 2024	Full text (open access)	IoT cleaning and power monitoring	Main proof of related work for cleaning
ii	Mukundaswamy et al., 2024	Abstract only	Cleaning plus water cooling via IoT	Closest published combination; read full text via college IEEE access
iii	Balamurugan and Nithya, 2025	Full text (journal copy)	Dry plus wet cleaning with IoT	Supporting evidence; see the correction below
iv	University of Sharjah patent, press release	Press release only	Thermal management	Patent-literature evidence that cooling matters

3.1 Source (i): IoT-based automatic cleaner and building energy

Citation. Anber M, Aboshosha BW. The impact of using an IoT-based automatic solar panel cleaner on building energy performance. International Journal of Low-Carbon Technologies 19 (2024) 1888-1900. DOI 10.1093/ijlct/ctae137. Open access under CC BY 4.0, so figures and tables may be reused with credit.

What they built. An ESP32 reads a voltage sensor and a current sensor, sends the data to a mobile dashboard, and starts a cleaning cycle when the reading falls more than 20 percent below the average. A DC motor turns the brush while stepper motors move it down and back along a threaded rod. Support parts are voltage regulators, a relay module and motor drivers. They drew the schematic in Fritzing.

What they measured. Two panels were exposed outdoors in Cairo for a year with measurements every two weeks, one with the cleaner and one without. Their panel was a 20 W polycrystalline module (450 x 350 mm, Vmp 18 V, Imp 1.11 A, Voc 21.96 V, Isc 1.16 A), the same class as yours. They report that dust reduced power by 89.33 percent, current by 40.3 percent and voltage by 11.8 percent, and that the cleaner restored 99.4 percent of the original power. A building-scale calculation then shows rooftop PV covering 10.5 percent of a 1,526,321 kWh annual demand.

Content you can reuse.

- The literature table of dust losses (cite the original studies, as in section 2).
- The comparison of manual, automatic and electrostatic cleaning on cost, efficiency, effort and scratch risk.
- The four-way component classification: input devices, output devices, controller, supporting parts.
- The efficiency-reduction formula and the 20 percent threshold idea.
- The recommendation for future work: drones, electrostatic and hydrophobic methods, and predictive maintenance with real-time data. Your digital twin and soiling index address the last of these.
- The recommendation to clean about every two weeks in a hot desert climate.

Caveats to state honestly. The abstract says the cleaner raised power by 89.33 percent, while the body uses the same number for the loss caused by dust; quote the body. Their Table 5 lists voltages up to about 48 V for a panel whose stated open-circuit voltage is 21.96 V, so the raw table is not internally consistent. Cite their summary conclusions, not their raw readings. The system has no cooling, no water use and no irradiance reference, so voltage or current alone cannot separate dust from clouds.

3.2 Source (ii): automated clean and cooling system using IoT

Citation. Mukundaswamy MS, Rajinikanth C, Shankarlingappa C. Automated Clean and Cooling System for Solar Photovoltaic Panels using IoT. 2024 4th International Conference on Data Engineering and Communication Systems (ICDECS), Bangalore, India. IEEE Xplore document 10502471.

What the abstract states. A microcontroller-based IoT system monitors the panel in real time and cleans it. Dust blocks sunlight and lowers efficiency. A cooling loop, with water circulating through tubes behind the panel, lowers the temperature, and efficiency improves as the operating temperature falls. The system covers dust monitoring, cleaning and cooling.

What I could not read. IEEE blocks automated access, so I have no component list, thresholds or results from the body. Read the full text through your college IEEE subscription and extract: the controller and sensors, the dust-detection method, the cooling flow rate and pump, the trigger temperatures, the measured power or efficiency gain, and the test duration. These fill the comparison table in section 5.

Why it matters. It is the closest published match to your idea of combined cleaning and cooling. Your differences are a front-surface spray that also washes the panel, a reference-cell soiling index, a cleaner-versus-control measurement mode, and net-energy accounting.

3.3 Source (iii): the link you gave and the paper you described

The link and your description do not match. The ResearchGate link returned an access error, so I found the same paper on the journal site. It is Balamurugan R, Nithya R, Advancements in solar panel maintenance: a review of IoT-integrated automatic dust cleaning systems, IAPGOS 15(1), 39-44, 2025, DOI 10.35784/iapgos.6614. It is an Arduino UNO system with an LDR, a voltage sensor, a Wi-Fi module, an LCD, relays, a water pump and a wiper motor. It does not use a camera, AI or servo motors.

The paper that does match camera and AI dust detection with servo-driven brushes is from the American International University-Bangladesh: A Real-Time IoT-Enabled Automated Solar Panel Cleaning System with Dust Detection Employing Advanced Image Processing Technique, 2024 ICAEEE, Gazipur (record). A camera and an NVIDIA Jetson judge dirtiness, and a brush driven by two servo motors sweeps across the panel until the camera sees it is clean. A related group paper reports a YOLOv5 model (82 percent classification accuracy, 15 W modules, about 39 percent average output gain after two sweeps of about 2 minutes) (record). Confirm which paper your teammate meant before citing.

Content from the link you gave.

- A two-stage clean: a dry brush for loose dust, then a wet pass with a mild vinegar and detergent solution.
- A light-gating idea: LDR and photodiode sensors prevent cleaning when light is low.
- Reported efficiency loss of 7.32, 10.78 and 14.52 percent after one day, one week and one month of dust.
- Four test cases (light and heavy sand, dry and wet cleaning); for example, output voltage rose from 9.4 V to 11.9 V after dry cleaning and to 14.4 V after the wet pass, against a clean-panel reading of 14.9 V.

- A cost comparison (about Rs 3,725 for their build) and cleaning twice a month instead of four times.

Caveats. Their Table 6 lists efficiencies of 21 to 24 percent for the cleaned panel, while Tables 2 to 4 show a clean-panel efficiency near 11 percent, so the headline 12.37 percent average improvement is not consistent with their own tables. Their text quotes current drops of 5.9 and 15.5 percent that also differ from the tables. Use the qualitative findings and the day, week and month loss figures with care, and say so in your report.

3.4 Source (iv): waste-air cooling patent from the University of Sharjah

What it is. A press release dated 6 August 2025 announces US patent US12341471B2 (listed on Google Patents), whose lead inventor is Prof. Chaouki Ghenai. The patent uses hot waste air from a central air-conditioning system to cool the rear surface of solar panels. An exhaust fan on the AC outlet blows air at panels held on a supporting structure at a set distance and tilt, with the fan calibrated for a target air temperature. The B2 kind code on the number indicates a granted patent; confirm on Google Patents.

Figures the inventors quote (not independently verified). Panels in hot regions can reach 70 °C and lose up to about 20 percent of output; each degree costs about 0.45 percent; every 10 °C rise is said to double the degradation rate; the system recovers up to 10 percent more power. The inventors also name the two challenges your project targets: thermal degradation and soiling.

Caveats. This is a press release, not the patent, so read the claims on Google Patents before citing any of it as a patent claim. It is HVAC-dependent rear cooling, so it is a different technique from your front water spray; treat it as evidence that temperature management is worth protecting, not as a competitor to copy.

3.5 How to cite these sources in your report

- Paraphrase and cite; keep quotations short and rare.
- Cite figures as "reported by" the authors, especially where the tables are inconsistent.
- Source (i) is open access (CC BY 4.0), so you may reproduce a figure with attribution. Treat the others as all-rights-reserved and redraw instead.

4. Related work, part 2: further sources

These sources back the four parts of SolarGuard: combined cleaning and cooling, water-spray cooling, soiling detection and scheduling, and digital twins. I read Sundah et al. in full; for every other row I read the abstract or summary page only, so confirm the details in the full text before quoting numbers.

4.1 Combined IoT cleaning and cooling systems

Source	What it shows	Use it for
Sundah et al., IJCA 186(53), Nov 2024	ESP32 with Blynk, voltage and PZEM sensors, an LDR and limit switches. Logic: if voltage drops below 15 V and light is above 400 lux, then cool if the panel is above 30 °C, otherwise clean. Cooling uses fans; only functional tests are reported, with no measured gain	Closest design logic to yours; shows the gap (no quantified net gain, light gating by LDR rather than a reference cell)
Mukundaswamy et al., ICDECS 2024	IoT cleaning, monitoring and a water-tube cooling loop behind the panel (abstract only)	Combined cleaning and cooling precedent
Febiola et al., Universitas Negeri Padang	ESP32 cooling system with Blynk monitoring and automatic logging to Google Sheets	ESP32 plus Blynk plus Sheets logging precedent
Smart temperature-dependent cooling using Arduino, IEEE 2023	Arduino-based automatic temperature-triggered cooling with optional IoT	Temperature-threshold cooling precedent

4.2 Water-spray and front-surface cooling experiments

Source	What it shows	Use it for
PLOS ONE 2024, comparison of cooling methods	Front water spray gave the best result: +3.9% efficiency versus no cooling, against +1.1% for air, +1.9% for water tubes behind the panel and +2.7% for a nanofluid	Justifies choosing a front spray over rear tubes
Spray on both panel sides (CROSBI 212860)	Total power gain up to 16.3% but effective gain 7.7% after the pump's own consumption; panel cooled from about 54 °C to 24 °C	The key precedent for net-gain accounting
Spray-cooling preprint, Jordan (Research Square)	Two identical panels tested together; surface temperature about 18.5% lower, peak power +15.6%; best when the spray hits the panel perpendicularly. Not peer reviewed	A/B test design and nozzle angle (treat as preliminary)
Rahim et al., 2021	Front spray at 5.80 L/min raised output to 253 W against 223 W for the reference panel	Shows a high flow rate; you will use far less water
Dorobantu et al., front water cooling	A perforated tube at the top lets water flow over the panel, reducing reflection loss and cleaning it at once	Supports the fixed top spray bar
Al-Azhar University dual-use system, reported by pv magazine, 2025	Cooling plus cleaning of bird droppings, with a 50 L reservoir and a collector at the bottom to recycle water; power gain 12 to 33% over a soiled panel	Water recycling idea and dual-use argument

4.3 Soiling detection, scheduling and patents

Source	What it shows	Use it for
US10607162B2	Two reference cells give a soiling level and rate; cleaning is scheduled when expected benefit exceeds cost	Prior art for the reference-cell index and economic trigger; read before claiming novelty
US9906190B2	Compares short-circuit currents of a soiled panel and a reference panel; forecasts a cleaning schedule	Prior art for your reference-cell method
EP4241315A1	Autonomous cleaning robot with predictive cleaning and an IoT dashboard	Prior art for robotic cleaning
NoSoilPV project (CORDIS)	Real-time soiling quantification without extra hardware and cleaning-time prediction	Soiling-estimation literature
Detection of cleaning interventions with machine learning, Applied Energy 2020	Detects cleaning events from temperature, voltage and current windows	Data-driven soiling analysis
AIUB camera and Jetson dust detection, ICAEEE 2024	Camera-based dust detection with servo-driven brush	Camera and AI alternative to your soiling index
Springer review of IoT and machine-learning dust cleaning	Reviews IoT and ML approaches to PV cleaning and the research gaps	Overview citation

4.4 Digital twins for photovoltaics

Source	What it shows	Use it for
Fadel and Alelaj, IJPEDS 16(3), 2025	Physics plus AI digital twin of a 250 W PV system in MATLAB/Simulink, RMSE 2.8 W, self-correcting	Definition and validation approach for a PV twin
Castilho et al., ISES Solar World Congress 2025	Open-source twin platform with real-time data, a Python simulation core, anomaly detection and dashboards	Architecture reference
Angelova et al., review of digital twins in PV (preprint, not peer reviewed)	Surveys twin concepts and PV applications	Background citation
Digital-twin approach for monitoring and event detection in PV systems (NSF PAR)	Compares measured output with twin predictions to detect faults	Fault-detection idea

4.5 What the evidence supports

- Dust loss is real and varies widely with site and time (section 2).
- Front water spray gives a larger cooling gain than rear tubes or air, and also washes the glass (section 4.2).
- The one study that subtracts pump power shows the effective gain can be roughly half the gross gain, so net accounting is essential.
- A reference-cell soiling index is established prior art, so your contribution is how it is combined, not the index itself (section 5).

5. Research gap and novelty statement

No source I read both cleans and cools a panel and reports whether the pump and motor energy is repaid by the extra generation. That accounting, tied to a measured soiling index and a built-in control panel, is the gap SolarGuard is designed to fill.

5.1 Feature comparison

"Not stated" means the part I could read does not say; it does not mean the feature is absent.

Capability	Anber 2024 (i)	Mukundaswamy 2024 (ii)	Balamurugan 2025 (iii)	Sundah 2024	Sharjah patent (iv)	SolarGuard
Automatic cleaning carriage	Yes	Yes	Yes	Yes	No	Yes
Active cooling	No	Yes (rear water tubes)	No	Yes (fans)	Yes (waste air, rear)	Yes (front spray)
Water spray also washes the panel	No	Not stated	Yes (wet pass)	No	No	Yes
Light or irradiance gating of the cleaning decision	No	Not stated	Yes (LDR)	Yes (LDR, 400 lux)	Not applicable	Yes (reference cell)
Soiling index independent of cloud and heat	No	Not stated	No	No	Not applicable	Yes
Untouched control panel	Yes	Not stated	Yes (dusty versus clean set)	Not stated	Not stated	Yes (Panel B)
Measures A and B at identical load	Not stated	Not stated	Not stated	Not stated	Not stated	Yes (relay K2)
Net energy gain after pump and motor	No	Not stated	No	No	Not stated	Yes (planned)
Digital twin or simulator	No	Not stated	No	No	No	Yes
Camera or AI dust detection	No	Not stated	No	No	No	Optional extension

5.2 Research questions

1. How much extra energy does front-spray cooling add to Panel A compared with the untouched Panel B at equal load?
2. How closely does one cleaning cycle restore Panel A to the clean-panel output, and how fast does the soiling index fall again?

3. After subtracting the pump and motor energy, is the net gain positive, and for which temperature and dust conditions?
4. Does an irradiance-normalised trigger clean less often than a fixed schedule for the same average output?

5.3 Candidate contributions

These are the points to defend in the paper. Items 1 and 3 overlap with prior art, so frame the contribution as the specific combination, and confirm it with a patent search first.

1. **A gated dual-function controller.** One controller uses a reference-cell soiling index and panel temperature thresholds to choose between cleaning and cooling, with rain, tank and jam interlocks. Related prior art: the decision logic in Sundah et al. and the reference-cell indices in US10607162B2 and US9906190B2.
2. **Automatic like-for-like measurement.** A relay periodically puts the cleaned and control panels on identical dummy loads so the logged comparison is fair.
3. **Net-energy accounting in firmware.** The controller logs extra generation, pump energy, motor energy and water used. The only similar accounting I found is the spray-cooling study that reports an effective gain.
4. **A twin-guided design.** Thresholds (45 °C on, 38 °C off, soiling index 0.90) are tuned in the digital twin before hardware, then checked against measured data.

5.4 Cautions for the patent plan

- Search Google Patents, Espacenet and the Indian patent database for the combinations above before filing.
- Do not publish the paper, post a video or present a poster before the application date; public disclosure can end novelty.
- Ask your university's innovation or IP cell. This document is not legal advice.

6. System architecture

SolarGuard has six layers: energy, sensing, control, actuation, mechanics and software. One ESP32-S2 sits in the middle, reads every sensor, and drives the pump and the cleaning motor from one shared battery rail.

Energy layer. Panel A feeds a charge controller, a 3-cell Li-ion pack with a BMS, a master switch and fuses. That produces the VBAT rail, 9.0 to 12.6 V, which feeds a buck converter (5.00 V for the logic), the pump branch and the motor branch. Panel B is the untouched

control and feeds a dummy load.

Sensing layer. Panel voltage and current for both panels, three DS18B20 temperature probes, a DHT22 for ambient air, a small reference solar cell as a sunlight meter, a rain probe, a tank level sensor and two carriage end stops.

Control layer. The ESP32-S2 runs a state machine (boot, homing, idle, cooling, cleaning, measure, fault) with safety interlocks and logs to a dashboard.

Actuation layer. Relay K1 switches the 12 V pump. An L298N drives the gear motor that moves the squeegee carriage. Relay K2 moves Panel A between the charger and a dummy load for fair A-versus-B comparison.

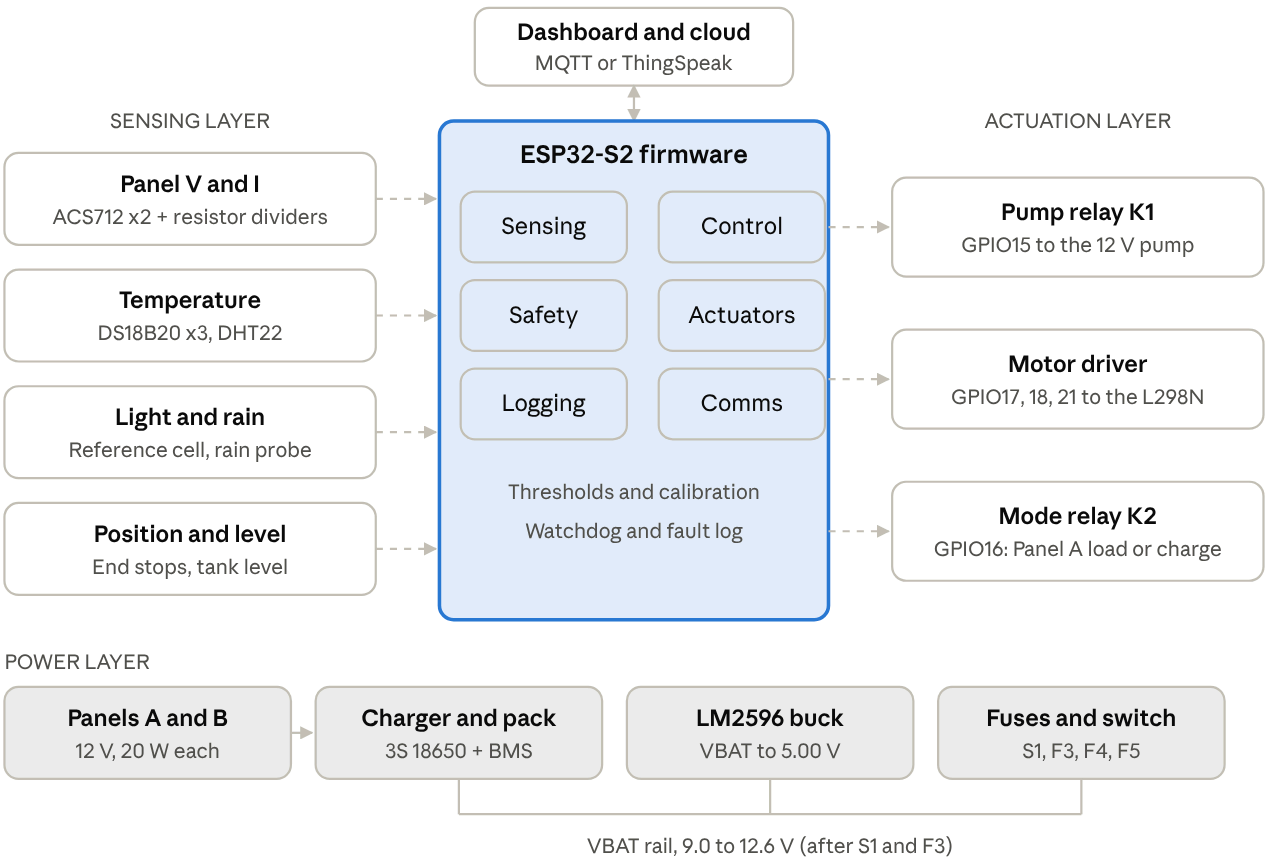
Mechanical layer. A fixed spray bar at the top, two 8 mm rods, bearing carriage blocks, a cross bar, wiper arms, a silicone squeegee, a GT2 belt, a motor and an idler.

Software layer. Firmware, Wi-Fi and NTP time, a dashboard, the browser digital twin and the Wokwi simulation.

6.1 Design principles

1. **One controller, simple parts.** Everything is an ESP32 plus off-the-shelf modules, so a first-year team can build and debug it.
2. **Always compare with a control.** Without Panel B, no claim about gain is defensible.
3. **Gate every decision.** Cleaning needs irradiance above a minimum and no rain; water use needs water in the tank and no rain.
4. **Fail safe.** Any fault turns the pump and motor off and logs a code.
5. **Simulate first.** Thresholds are tried in the digital twin and the Wokwi simulation before any hardware runs.

The figure below shows how the layers connect; solid lines carry energy and dashed lines carry signals.



system architecture · sensing, firmware, actuation and power layers

7. Components and the function of each part

Reference designators (U, R, C, D, F, K, S, M, P, PV) are used in every diagram and netlist in this report. The Source column says whether the part is in the college labs (checked against the lab lists you supplied; confirm quantities with the lab staff) or must be bought. Prices were not verified.

7.1 Power and protection

Ref	Part	Qty	Function	Source
PV1	12 V, 20 W solar panel (about 18 V at maximum power, 22 V open circuit, 1.2 A short circuit)	1	Test panel: converts sunlight to DC; it is the one that is cleaned and cooled	Buy
PV2	Identical 12 V, 20 W panel	1	Control panel: left untouched so every gain can be measured against it	Buy
PV3	Reference mini solar cell (3 V, 150 mA, 0.45 W)	1	Its current is proportional to sunlight, so it tells dust from clouds	Lab (AB3-611)
U1	Solar charge controller with a Li-ion 3S setting, or a CN3791 module set to 3S	1	Charges the pack from Panel A without overcharging it	Buy
BT1-BT3	18650 Li-ion cell, 2600 mAh	3	Store about 28.9 Wh for the electronics, pump and motor	Lab
U2	3S BMS, 10-20 A	1	Protects the cells from overcharge, over-discharge and short circuit, and balances them	Buy
-	3S 18650 holder (or two 2-cell holders with one slot bridged)	1	Holds the three cells	Lab (2-cell) or buy
U3	LM2596 buck converter module	1	Steps VBAT (9.0-12.6 V) down to a steady 5.00 V for the logic	Lab
D1	Schottky diode, SR560 or 1N5822 (3 A or more)	1	Stops current flowing back from the battery into the panel at night	Buy
D2	1N4007 diode	1	Flyback diode across the pump; absorbs the voltage spike when the pump switches off	Lab

Ref	Part	Qty	Function	Source
F1, F2	3 A fuse and holder	2	Protect the panel lines and keep a fault from over-driving the ADC pins	Buy
F3	5 A fuse and holder	1	Main fuse on the VBAT rail	Buy
F4, F5	2 A fuse and holder	2	Protect the pump and motor-driver branches	Buy
S1	Master switch rated at least 6 A DC	1	Cuts all loads at once	Buy
RL1, RL2	15 ohm, 25 W power resistor	2	Identical dummy loads so Panel A and Panel B are compared at the same operating point	Buy

7.2 Controller and sensing

Ref	Part	Qty	Function	Source
U6	ESP32-S2-DevKitM-1	1	The controller: reads sensors, runs the state machine, drives outputs, uses Wi-Fi for logging and time	Lab
U4, U5	ACS712-30A current sensor module	2	Hall-effect sensors that measure the current of Panel A and Panel B	Lab (AB1-213)
R1-R4, R5-R8	Resistors 100k, 10k and 4.7k for the voltage and current dividers	-	Scale panel voltage and sensor output into the ESP32's safe analog range	Lab (AB3-310)
C4-C7	100 nF ceramic capacitor	4	Filter noise on the four analog inputs	Lab (AB3-310)
U7, U8, U9	DS18B20 waterproof temperature probe	3	Panel A back, Panel B back and tank water or shade temperature	Lab (AB3-310)

Ref	Part	Qty	Function	Source
U10	DHT22 temperature and humidity sensor	1	Ambient air condition	Lab
R9	10 ohm, 0.5 W resistor	1	Turns the reference cell's current into a readable voltage	Lab
U11	Soil-moisture module used as a rain probe	1	Detects rain so water use is skipped	Lab
U12	Optical liquid-level sensor (or ultrasonic module)	1	Tells the controller the tank has water; protects the pump from running dry	Lab (AB3-611)
U13, U14	IR sensor modules or push buttons	2	Top and bottom end stops for the carriage	Lab
U17	AMG8833 or MLX90614 thermal sensor	1	Optional hotspot or non-contact temperature reading	Lab
U18	ESP32-CAM (OV3660)	1	Optional dust photos for later image-based detection	Lab

7.3 Actuation

Ref	Part	Qty	Function	Source
U15 (K1, K2)	5 V, 2-channel relay module	1	K1 switches the pump; K2 moves Panel A between the charger and the dummy load	Lab
U16	L298N dual H-bridge motor driver	1	Sets the direction and speed of the carriage motor	Lab
M1	25GA-370 12 V gear motor, 60 RPM	1	Drives the GT2 belt that moves the squeegee carriage	Lab
P1	12 V mini diaphragm pump, 2-4 L/min	1	Pushes water to the spray bar for cooling and washing	Buy

7.4 Water system and mechanics

Part	Qty	Function	Source
Misting nozzles	5	Spread a fine water film over the panel	Buy
12 mm tube (spray bar)	1	Holds the nozzles across the top edge	Buy
Silicone tubing, check valve, T-joints	1 set	Plumbing from tank and pump to the spray bar	Buy
Inline filter or mesh	1	Keeps particles out of the nozzles	Buy
Water tank, 2-5 L (lab container box if watertight)	1	Reservoir	Lab or buy
8 mm smooth rods	2	Guide rails for the carriage	Buy
LM8UU linear bearings	4	Let the carriage blocks slide on the rods	Buy
8 mm aluminium cross bar	1	Joins the two carriage blocks and carries the wiper arms	Buy
15 x 15 mm aluminium angle and a flat strip	1 set	Backbone that clamps the silicone squeegee	Buy
Silicone squeegee strip	1	The wiping edge	Buy
GT2 belt, 20-tooth pulley, 20-tooth idler	1 set	Converts motor rotation into carriage travel	Buy
Springs or rubber bands	2	Press the squeegee gently on the glass	Buy
Tilt stand (aluminium angle or wood) and fasteners	1 set	Holds both panels at a tilt	Buy
3D printed parts (section 12)	12 types	Blocks, arms, brackets, mounts	Print

7.5 Electrical supplies and enclosure

Part	Qty	Function	Source
Capacitors: 220 uF and 470 uF electrolytic	2	Smooth the motor supply (C1) and the 5 V rail (C3)	Lab (220 uF) or buy
Power wire 1.0-1.5 mm2, hookup wire 22-24 AWG	5 m each	Power and signal wiring	Buy
Screw terminals, XT60 connectors, perfboard	1 set	Reliable permanent connections instead of a breadboard	Buy
Steel box or IP65 enclosure with cable glands	1	Protects the electronics; keep it above the splash zone	Lab (steel box) or buy
Heat shrink, cable ties, tape	1 set	Finishing and strain relief	Buy

7.6 Tools

Multimeter, soldering iron with solder, flux and a desoldering pump, wire stripper, pliers and tweezers, jumper wires and a breadboard (for testing only) are all available in the labs. Buy a screwdriver set and a hot glue gun if you do not have them.